Primary Key (Column Based)

- All activities we do with other keys.
- Always → Not Null
  → Unique.
- Each table has only 1 primary key.

```
CREATE TABLE Customer
(id INT PRIMARY KEY,
 branch_id INT,
 first_name CHAR (90),
 last_name CHAR (50),
 date_of_birth DATE
 gender CHAR (6),
 PRIMARY KEY (id)
```

FOREIGN KEY (FK)

Customer → Account

Each table can have any no. of FK.

```
CREATE TABLE Account ( id INT,
                       balance CHAR (50),
                       customer_id INT, PRIMARY KEY (id),
                                        FOREIGN KEY (Custo
```

mer_id)

REFERENCES (Customer (id))

SQL Constraints

NOT NULL

PRIMARY KEY

FOREIGN KEY

UNIQUE

CHECK

DEFAULT

PK → To be unique, Not NULL

UNIQUE → To be unique
- when there is no PK
- Could be null
- can have any no. of unique keys.

```
CREATE TABLE Customer (customer_id INT UNIQUE,
                       phone_model INT UNIQUE);
```

Syntax

```
CONSTRAINT const_name CHECK (expr);
```

```
CREATE TABLE ACCOUNT (Account_id INT
                      Account_Balance DOUBLE,
                      CONSTRAINT acc_balance_check
                      CHECK (account_balance >=500))
```

user defined

```
CREATE TABLE loan (
    loan_id INT,
    loan_interest_rate DOUBLE NOT NULL
    loan_type_id INT
    )
```

```
CREATE TABLE People (
    PID INT PRIMARY KEY,
    LastName VARCHAR (255),
    FirstName VARCHAR (255);
    Address VARCHAR (255),
    City VARCHAR (255)
);
```

```
CREATE TABLE Account (
    id INT,
    customer_id INT,
    balance CHAR (50) NOT NULL,
    Primary Key (id),
    foreign key (customer_id) References Customer (id), on this
    Foreign key (card_id) References
    Card (id)
    Constraint balance_check
    CHECK (balance > 500)
```

because we are applying check constraint

→ Data Definition language

ALTER OPERATIONS

↓

To change structure of the data.

CREATE TABLE (ver_0)

(ver_0)

A_id	A_no.	A_Type

(ver_1) (ADD)

(To add to new column)

A_id	A_num	A_type	A_balance

To change the column name

Account_id	A_num		

ver_2 (CHANGE COLUMN)

- To modify something in existing table → eg changing datatype from VARCHAR → CHAR MODIFY

Account_id	A_num	A_type	A_balance

Account_id	A_num	A_type

Delete column eg DROP COLUMN

ALTER TABLE WITH ADD

ALTER TABLE table_name

ADD new_col_name datatype,
FIRST A_id; (Adding before A_id)
ADD new_col_name2 datatype;
AFTER A_id (Added after A_id)

- if we want it after something

MODIFY

ALTER TABLE table_name

MODIFY colname col_datatype;

DESC Insurance
↓
To know structure of table.

CHANGE COLUMN

```
ALTER TABLE table_name
        CHANGE COLUMN old_col_name
                new_col_name datatype;
```

ALTER TABLE WITH DROP

```
ALTER TABLE table_name
        DROP COLUMN col_name;
```

(no need to mention datatype)

ALTER TABLE RENAME (Renaming Table)

```
ALTER TABLE Insurance
        RENAME TO Insurance_temp;
```